# Problem Set 3

Quan Wen

Due: 19 Dec 2025

## Ring network

Consider the ring network model we discussed in the class

$$\tau \frac{du_i}{dt} = -u_i + F\left(\sum_{j=1..N} w_{ij} u_j + I_i^0\right), \tag{1}$$

where $F(x) = [x]_+$ is a rectified linear function and the weights between the neurons are determined by their angular difference and thus are translational invariant.

$$w_{ij} = \frac{1}{N} J(\theta_i - \theta_j), \tag{2}$$

where $\theta_i = -\pi + \frac{2\pi i}{N}$ and $J$ is a $2\pi$ periodic function, and $I_i^0$ is also a periodic $2\pi$ function. In the continuous limit, we have

$$\tau \frac{\partial u(\theta,t)}{\partial t} = -u(\theta,t) + \left[\frac{1}{2\pi}\int_{-\pi}^{\pi} J(\theta-\theta')u(\theta',t)d\theta' + I^0(\theta)\right]_+, \tag{3}$$

(a) If the connectivity matrix $J$ is translational invariant and symmetric, prove that the eigenfunctions of $J$ are Fourier basis, namely

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} d\theta' J(\theta-\theta')e_\mu(\theta') = \lambda_\mu e_\mu(\theta) \tag{4}$$

$$e_\mu \sim \cos(\mu\theta) + \sin(\mu\theta), \tag{5}$$

where $\mu = 0, 1, 2, ....$ Find out the correct normalization factor for the eigenfunctions. The corresponding eigenvalues of the the connectivity matrix can be viewed as the Fourier transform $J(\theta)$, namely

$$\lambda_\mu = \frac{1}{2\pi}\int_{-\pi}^{\pi} J(\theta)\cos(\mu\theta)d\theta \tag{6}$$

(b) In the class, we consider a simplified connectivity $J$

$$J(\theta) = J_0 + J_1 \cos\theta \tag{7}$$

$$I_i^0(\theta) = I_0 + I_1 \cos(\theta - \theta_0) \tag{8}$$

Consider the special case $I_1 = 0$, so that each neuron receives the same homogeneous inputs and $F(x) = [x]_+$. We show in the class that a marginal stable solution is the emergence of a bump, namely, $u(\theta) = [u_1 \cos(\theta) + u_0]_+$. Please show that the bump can actually appear anywhere, namely $u(\theta) = [u_1 \cos(\theta - \phi) + u_0]_+$, where $\phi$ is arbitrary.

(c) A naive solution is that the population neural activity is homogeneous : all neurons have the same activity. Show that this naive solution with $u_1 = 0$ is unstable when $J_1 > 2$.

(d) Draw the phase diagram on the $J_0 - J_1$ plane. Discuss and plot in what region, the system is marginally stable? In what region, the system exhibits homogeneous activity? In what region, the system is unstable?

(e) Perform computer simulation to confirm your results using a discrete ring network with $N$ neurons for (a)-(c).

(f) Now consider some tiny modulatory input with $0 < I_1 \ll I_0$ and $J_1 > 2$. Show that in this case, the location of the peak of the stationary activity is not arbitrary, but aligned to the stimulus angle $\theta_0$.

(g*) Add to the connectivity matrix a term $\frac{J_1}{N}\gamma \sin(\theta_i - \theta_j)$, where $|\gamma| \ll 1$. Show that there is a solution with the form

$$u(\theta, t) = f(\theta - \omega t), \tag{9}$$

where $f(\theta)$ is the steady state activity profile calculated in the class for $\gamma = 0$ and the angular velocity satisfies

$$\omega = \frac{\gamma}{\tau} \tag{10}$$

**Here is some specific guide the (g*):**
A. Assume a traveling profile of the form of 9 (for now just use an arbitrary profile $f(\theta)$ ) and insert it into the two sides of 1. Express the RHS in terms of the order parameters as was done in the class. Expand the RHS in powers of $\gamma$ and keep only terms up to linear in $\gamma$.
B. Show that if $f(\theta)$ is the profile for $\gamma = 0$ the dynamic equations 1 are satisfied.

# Low Rank RNN

Consider a nonlinear RNN with random Gaussian coupling matrix:

$$\frac{dx_i}{dt} = -x_i + \sum_{j=1}^{N} J_{ij}\phi[x_j(t)],$$

with $\phi(x) = \tanh(x)$ and

$$\langle J_{ij} \rangle = 0, \qquad \langle J_{ij} J_{kl} \rangle = \frac{g^2}{N}\delta_{ik}\delta_{jl}.$$

Now we add a rank-one coupling term to the random couplings between neurons to impose some structure in the connectivity matrix:

$$J_{ij} = g\chi_{ij} + \frac{m_i n_j}{N}, \tag{11}$$

where $\chi_{ij}$ is the Gaussian random bulk matrix, whose entries have zero mean and variance $\frac{1}{N}$. The rank-one coupling is the external product between two one-dimensional vectors $\vec{m}$ and $\vec{n}$, whose entries are on the order of $O(1)$.

(a) Although the structural term is on the order of $O(1/N)$, The collective effect on the neural activity $x_i$ can be significant. In particular, show that the mean activity of $x_i$ is not diminishing, with

$$\mu_i = \kappa m_i,$$

where $\kappa$ is the alignment between the $\vec{n}$ and the average population firing rate:

$$\kappa = \frac{1}{N}\sum_{j=1}^{N} n_j \langle \phi_j \rangle.$$

In the last equation, we adopted the short-hand notation $\phi_j(t) := \phi[x_j(t)]$. Here $\langle \phi_j \rangle$ is the average firing rate of unit $j$, i.e., $\phi[x_j(t)]$ averaged over the Gaussian variable $x_j$.

(b) Follow my DMFT note compute the effective input correlations for this rank-one model.

(c) calculate the eigenvalues of the $J$ matrix and plot them on the complex plane. Describe what you have seen. How does the rank-one matrix modify the eigenspectrum?

(d) Perform computer simulation of the rank-one RNN network with $N = 1000$. $\vec{m}$ and $\vec{n}$ can be simulated as two random vectors, with the random coupling strength $g$ and low-rank coupling strength $\frac{m^T n}{N}$ as two tunable parameters. Explore the behavior of the system as you change the two parameters. In which regime the steady state of the system is quiescent? In which regime the system is chaotic? In which regime the system is bistable?